

# Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

## Justification

Data instances span scientific literature, clinical trials, clinical cases, drug leaflets, prescription transcripts, and exam questions [Q19], providing pharmaceutical and posology vocabulary across many datasets. However, for the regulatory deployment, the content lacks systematic coverage of INNs as distinct from brand names, ATC codes, excipient nomenclature, EMA posology templates, and contraindication qualifier phrases as targeted instances. QUAERO EMEA contains the most regulatory-adjacent content (QUAERO-D1 through QUAERO-D9) but its sentence-splitting and nested-entity simplification reduce fidelity [Q37]. Multiple datasets contain non-metropolitan French content (DiaMED Pan African, E3C Morocco/sub-Saharan Africa, MORFITT Egypt/Saudi/Jordan/Canada) that diverges from the deployment's primary metropolitan-France scope. French overseas territory content and tropical pathology vocabulary are absent except for incidental mentions (MORFITT-D7, CAS-D18).

## Strengths

- Drug names (INNs and brand names), dosage, route, and frequency vocabulary appears densely across QUAERO, PxCorpus, DEFT2021, CAS, and ESSAI (QUAERO-D7, PXCORPUS-D2, DEFT2021-D6, CAS-D7, ESSAI-D5)
- Drug safety, contraindication, and adverse event content present (QUAERO-D5, QUAERO-D6, DEFT2020-D2, DEFT2020-D3, FRENCHMEDMCQA-D6)
- QUAERO EMEA provides authentic SmPC composition sections, EU marketing authorization references, and excipient lists with E-numbers (QUAERO-D1, QUAERO-D2, QUAERO-D9)
- Formal written French biomedical register consistent across all datasets (CAS-D1, ESSAI-D11, QUAERO-D9)

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                                        |
|------|----------------------------------------------------------------------------|
| IC-1 | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| IC-2 | Does culturally sensitive content align with target culture?               |
| IC-3 | Flag inputs assuming Western-specific knowledge                            |
| IC-4 | Regional annotator recruitment for culturally sensitive instances          |
| IC-5 | Document content issues harming content validity                           |

### Checklist responses:

IC-1: The deployment requires region-specific knowledge of EU/French regulatory vocabulary (INNs, ATC codes, EMA QRD template phrasing, ANSM blue-box content, French List I/II classification) [WEB-5, WEB-6]. The benchmark provides partial vocabulary (drug names, dosages) but lacks systematic regulatory-template phrasing. ANSM-specific France packaging requirements (CIP codes, Exploitant designation) are not represented in benchmark instances.

| ID    | Evidence                                                                                                                                                                           |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-5 | ANSM France-specific submission requirements include Feuille de style template and blue-box for CIP codes                                                                          |
| WEB-6 | Variation Type IA/IB/II classification framework establishes legal significance of small text changes — regulatory annotators would score differently than clinical/NLP annotators |

|                  |                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QUAERO-D2</b> | TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab ( 20 mg / ml ) ; phosphate de sodium ... polysorbate 80 ( E433 ) et eau pour préparation injectable — Full SmPC composition section with INN, dosage, excipients |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

IC-2: Cultural/regulatory alignment with metropolitan France is partial. QUAERO EMEA aligns well (QUAERO-D9, QUAERO-D10). However, multiple datasets contain non-metropolitan content: DiaMED is Pan African (DIAMED-D9, DIAMED-D12); E3C includes Morocco and sub-Saharan Africa (E3C-D10, E3C-D11, E3C-D12); MORFITT includes Canadian, Egyptian, Saudi, Jordanian populations (MORFITT-D12 through MORFITT-D15); MANTRAGSC fr\_medline includes 1971 Cameroon and Senegal cases (MANTRAGSC-D6, MANTRAGSC-D10). This may dilute metropolitan-France signal.

| ID                | Evidence                                                                                                                                                                                                                                       |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-1</b>      | French overseas territories (Martinique, Guadeloupe, French Guiana, Réunion, Mayotte) are DOM under Article 73 — integral parts of the French Republic                                                                                         |
| <b>DIAMED-D9</b>  | service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabé institution; confirms non-metropolitan-France provenance                                                                                  |
| <b>E3C-D10</b>    | Monsieur F M âgé de 70 ans, originaire du Maroc — Patient origin stated as Morocco; suggests non-metropolitan French clinical context                                                                                                          |
| <b>MORFITT-D7</b> | l'étude de l'épidémie de Chikungunya, un alphavirus transmis par Aedes aegypti et Aedes albopictus, survenue dans l'océan Indien en 2004-2007. — Tropical arboviral disease content; incidental coverage of French overseas territory diseases |

IC-3: Western/Francophone-Africa-specific content is present but the orientation is non-uniform. Some content (FrenchMedMCQA nuclear physics, beekeeping STS pairs DEFT2020-D11) is irrelevant. No tropical pathology vocabulary calibrated to French DOM territories (dengue, chikungunya, paludisme, leptospirose, leishmaniose) is systematically represented despite being identified as the secondary deployment scope [WEB-1].

| ID                  | Evidence                                                                                                                                                                                                                                       |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-1</b>        | French overseas territories (Martinique, Guadeloupe, French Guiana, Réunion, Mayotte) are DOM under Article 73 — integral parts of the French Republic                                                                                         |
| <b>DEFT2020-D11</b> | Certains apiculteurs sélectionnent leurs reines afin de favoriser au mieux la production. — Beekeeping topic — irrelevant to regulatory deployment                                                                                             |
| <b>MORFITT-D7</b>   | l'étude de l'épidémie de Chikungunya, un alphavirus transmis par Aedes aegypti et Aedes albopictus, survenue dans l'océan Indien en 2004-2007. — Tropical arboviral disease content; incidental coverage of French overseas territory diseases |
| <b>CAS-D18</b>      | une sérologie leptospirose devient positive ... trois semaines après le début des symptômes . — Leptospirosis serology; pathology with elevated prevalence in tropical/overseas contexts                                                       |

IC-4: INSUFFICIENT DOCUMENTATION on regional annotator recruitment for content review. Benchmark documentation does not describe content sensitivity review by regulatory affairs experts. Would need expert elicitation with regulatory affairs specialists to systematically identify culturally/regulatorily sensitive instances.

IC-5: Significant content validity issues: (a) regulatory entity vocabulary (ATC codes, INN-specific labels) untagged or absent (QUAERO-D11, QUAERO-D14); (b) non-metropolitan content dilutes France-specific signal; (c) overseas territory tropical pathology absent; (d) document fragmentation artifacts in QUAERO (QUAERO-D15, QUAERO-D16, QUAERO-D17) reduce instance quality.

| ID                | Evidence                                                                                                                                                                                                                                                   |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QUAERO-D11</b> | Agent antithrombotique – inhibiteur direct de la thrombine , code ATC : La lépirudine ... est une hirudine recombinante — ATC code string unlabeled (tag=0); INN and endogenous enzyme both CHEM — scheme collapses regulatory-legally distinct categories |
| <b>QUAERO-D15</b> | o . — Single-character fragment artifact from document splitting                                                                                                                                                                                           |

## Information Gaps

- MANTRAGSC fr\_emea subset content not directly sampled
- Whether FrenchMedMCQA pharmacy-exam content includes pharmacovigilance or regulatory-affairs sub-domains

## Requires Expert Verification

- Whether ANSM-specific safety warning content phrasing diverges from EMA QRD in ways the benchmark fails to represent

- Sensitivity review of non-metropolitan content for downstream model bias
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## Remediation

### **Gap 1: Tropical pathology vocabulary and French overseas territory content absent despite secondary deployment scope (Martinique, Guadeloupe, French Guiana, Réunion, Mayotte)**

Recommendation: Construct a tropical pathology supplementation set with dengue, chikungunya, paludisme, leptospirose, leishmaniose vocabulary from French DOM clinical and regulatory sources. Cross-reference with ANSM regional safety communications. Validate model performance separately on metropolitan and DOM evaluation splits.